

OMKARNATH THAKUR

📞 602-668-0832 ✉ othakur@umd.edu 💻 [info-stats-ai](https://info-stats-ai.github.io) 🌐 github.com/Info-stats-ai

Education

University of Maryland, College Park

Master of Science in Data Science

Jan 2024 – Dec 2025

College Park, Maryland

Technical Skills

Languages: Python, SQL, R, Java

Backend/Frontend: FastAPI, Flask, Node.js, Express.js, Next.js, HTML, JavaScript/TypeScript, MongoDB

Pipelines: Machine Learning, Deep Learning, Generative AI, Natural Language Processing, RAG, Agents, Prompt Engineering

Frameworks: PyTorch, Pandas, NLTK, TF-IDF, spaCy, NumPy, Scikit-learn, TensorFlow, KerasTuner, LangChain

AI Tools: LangGraph, CrewAI, Hugging Face, AgentSDK

Deployment & Monitoring: MCP, Docker, Kubernetes, Grafana, GitHub Actions, Airflow, MLflow, Terraform, AWS, Azure, GCP

Training & Evaluation: Fine-tuning, PEFT, QLoRA, Knowledge Distillation, Mixture of Experts, Cost Optimization, vLLM

Experience

Machine Learning Engineer

Feb 2026 – Present

Community Dreams Foundation

- Architected an end-to-end research data platform using **Python, SQL, and AWS**, ingesting longitudinal career data from external APIs into **S3 bronze/silver/gold** layers and enabling scalable analytics through **AWS Glue** and **Amazon Athena**.
- Designed **SQL**-driven analytical pipelines in **Athena** to construct career trajectory and transition datasets, improving reproducibility and usability for downstream analysis of time-evolving structured data.
- Extracted role, event, and entrepreneurship signals from structured tables and unstructured text using data mining, machine learning, and **LLM**-based text extraction to enrich downstream research features.
- Optimized high-throughput **vLLM** inference with CUDA-aware batching, metadata tracking, and prompt versioning, improving run reproducibility while supporting efficient large-scale text processing.
- Scaled predictive model training with multi-GPU **PyTorch DistributedDataParallel (DDP)**, handling distributed sampling, synchronization, and checkpointing while logging experiments in **Weights & Biases** and **MLflow** for robust benchmarking.

Artificial Intelligence Intern

Jun 2025 – Aug 2025

The Builder Market

Los Angeles, CA

- Developed a production-grade **Agentic RAG** assistant using **MERN, PostgreSQL, LangChain, and OpenAI APIs** to answer construction workflow queries, improving response grounding through hierarchical chunking and reranking.
- Evaluated retrieval performance using **MRR, nDCG, and LLM-as-a-Judge**, tuning chunking and ranking strategies to improve answer relevance and overall retrieval quality.
- Engineered a large-scale ML pipeline processing **400K+** product descriptions across **8** categories using **Python, Hugging Face, and OpenAI APIs**, benchmarking **12** traditional ML and LLM-based models.
- Fine-tuned **GPT-4o-mini** on **JSONL** datasets with **Weights & Biases** tracking, achieving **70–80%** accuracy, **15–25%** error, and **RMSLE 0.3–0.5**, outperforming baselines by **40–50%**.

Data Scientist

Jun 2020 – Jul 2023

Kamdhenu Robotics

Bengaluru, India

- Engineered an end-to-end call summarization solution by fine-tuning a **BERT** model and automating deployment to **AWS SageMaker** through Dockerized **GitHub Actions** CI/CD, saving analysts **30 minutes** of reporting time per call.
- Optimized experimentation efficiency by developing a **K-Means** clustering pipeline for time-series peak detection and applying **ARIMA** and regression models for forecasting, cutting redundant runs and saving approximately **\$50K annually**.
- Developed a transformer-based **Named Entity Recognition (NER)** pipeline with **80%** accuracy to extract vendors, products, and markets from call transcripts, improving insight generation from unstructured business text.
- Orchestrated a multi-modal AI pipeline with **FastAPI** to process PDFs, images, CSVs, PPTs, and raw text, converting heterogeneous inputs into usable text and audio outputs through a unified workflow.
- Deployed a **RAG** pipeline with **LoRA**-tuned **Mistral-7B**, enabling grounded responses and improved answer quality across structured and unstructured data sources.

Projects

OpenTSLM Paper Implementation | Python, PyTorch, Hugging Face, CUDA, Llama, Gemma

2025

- Reproduced **OpenTSLM** by extending pretrained **Llama/Gemma** models for reasoning over multivariate time-series data.
- Benchmarked training/evaluation across HAR, SleepEDF, ECGQA, M4 using curriculum and multimodal inference workflows.
- Validated pretrained checkpoints with **CUDA**-based inference and structured benchmarking to improve experimentation efficiency.

Curious RL Paper Implementation | Python, PyTorch, Reinforcement Learning, A2C, VizDoom

2025

- Reproduced exploration paper by integrating an **Intrinsic Curiosity Module** with the **A2C** reinforcement algorithm in Python.
- Trained/evaluated agents on **VizDoom** to replicate exploration behavior and benchmark performance against vanilla **A2C**.
- Analyzed training stability with **TensorBoard** monitored and regulated, identified local-optima failure modes and agent learning.